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College Name: SOCET

Program: B-TECH

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GitHub link: https://github.com/Yuvraj2709/PP_MiniProject

by Yuvraj Pithwa

Advanced Python Libraries

Unlocking Data Analysis and Visualization

Intro to Python Libraries

Python Libraries

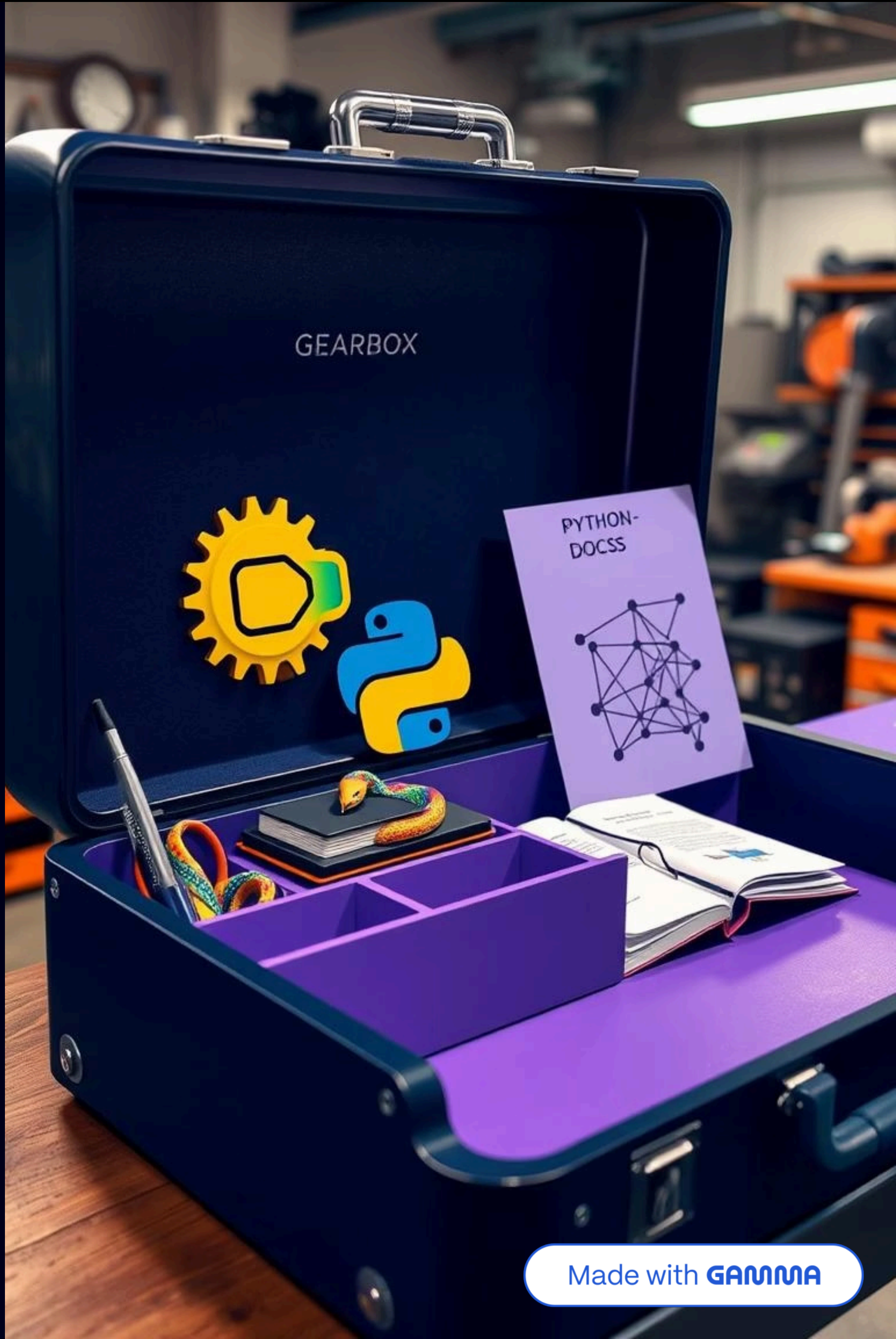
Collections of reusable code

Simplify Tasks

Make code modular and reusable

Advanced Libraries

NumPy, Pandas, Matplotlib



NumPy Overview

Numerical Computing

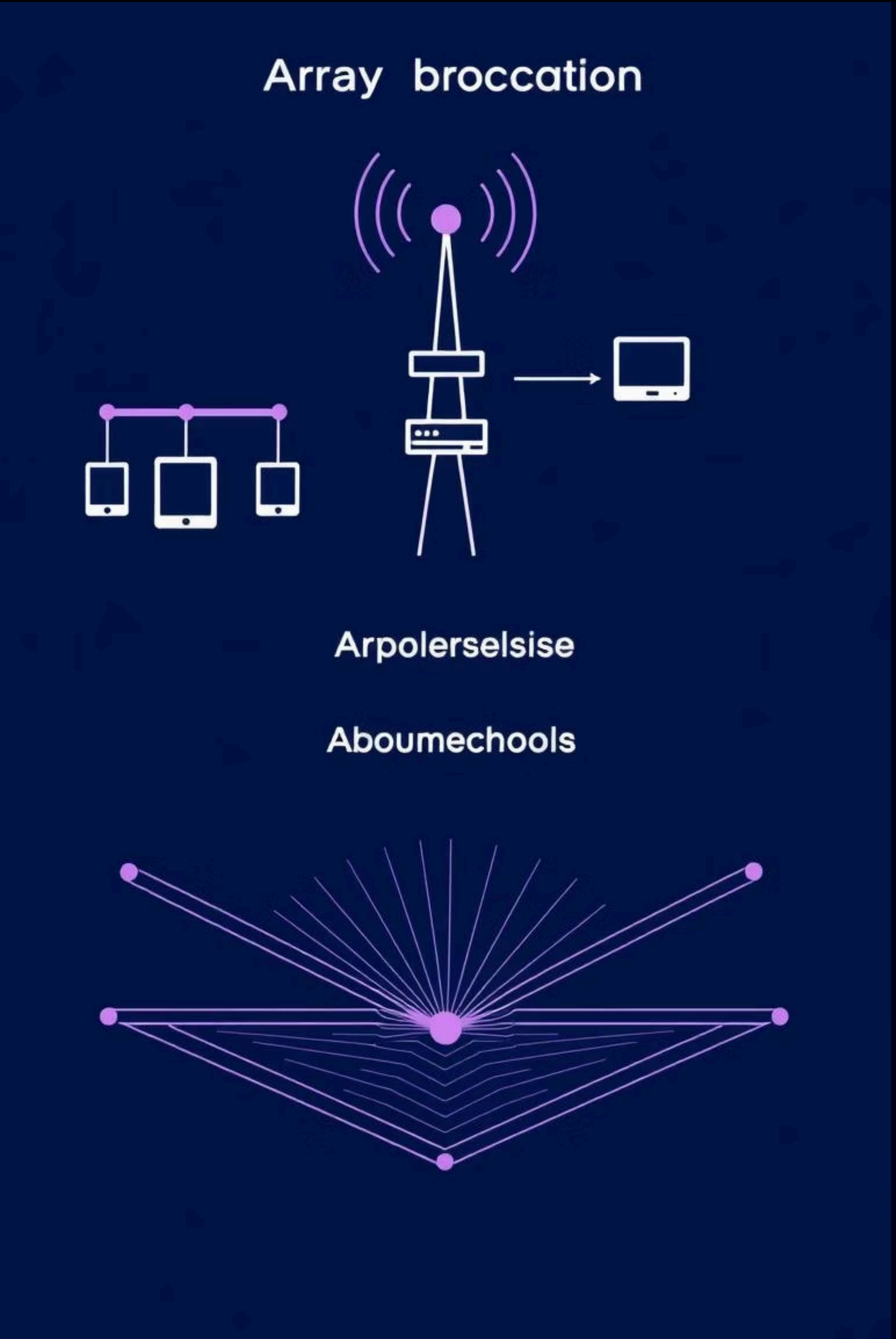
Backbone for scientific tasks

Multi-Dimensional Arrays

Efficient storage and handling

Mathematical Operations

Essential scientific functions



Key NumPy Features

Multi-Dimensional Arrays

Broadcasting

Vectorization

Linear Algebra

Applications of NumPy

NumPy provides powerful tools for image manipulation and analysis.

- **Image Representation:** Images can be represented as multi-dimensional NumPy arrays, where pixel values correspond to array elements.
- **Image Processing:** NumPy enables various image processing operations such as filtering, enhancement, and transformations.
- **Image Analysis:** NumPy facilitates tasks like object detection, feature extraction, and image segmentation.

Pandas Overview

1

Data Manipulation

Powerful data handling

2

Data Analysis Library

Structured data analysis

3

DataFrames and Series

Easy-to-use data structures

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10017	624	662	1999	12207	044
21985	094	007	1798	12806	098
19055	035	068	1990	12397	099
19075	334	001	1373	12301	290
19962	905	007	1635	12905	062
11900	699	1310	1192	12330	085
11992	746	1210	1903	12248	064
19909	067	1280	1738	12394	088
23061	096	1210	1770	12427	034
27366	060	1230	1175	12395	096
19981	057	1281	1258	12345	7.08
19915	1394	1288	1248	12140	494
19955	1216	1253	1596	12245	454
19954	1947	1286	1495	12346	34
19083	1137	1100	1577	1	30
19961	1313	1575	4.10		
19962	1977	1577	000		
1780	1745	1795	000		
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Key Pandas Features

Data Cleaning

Handle missing values

Transformation

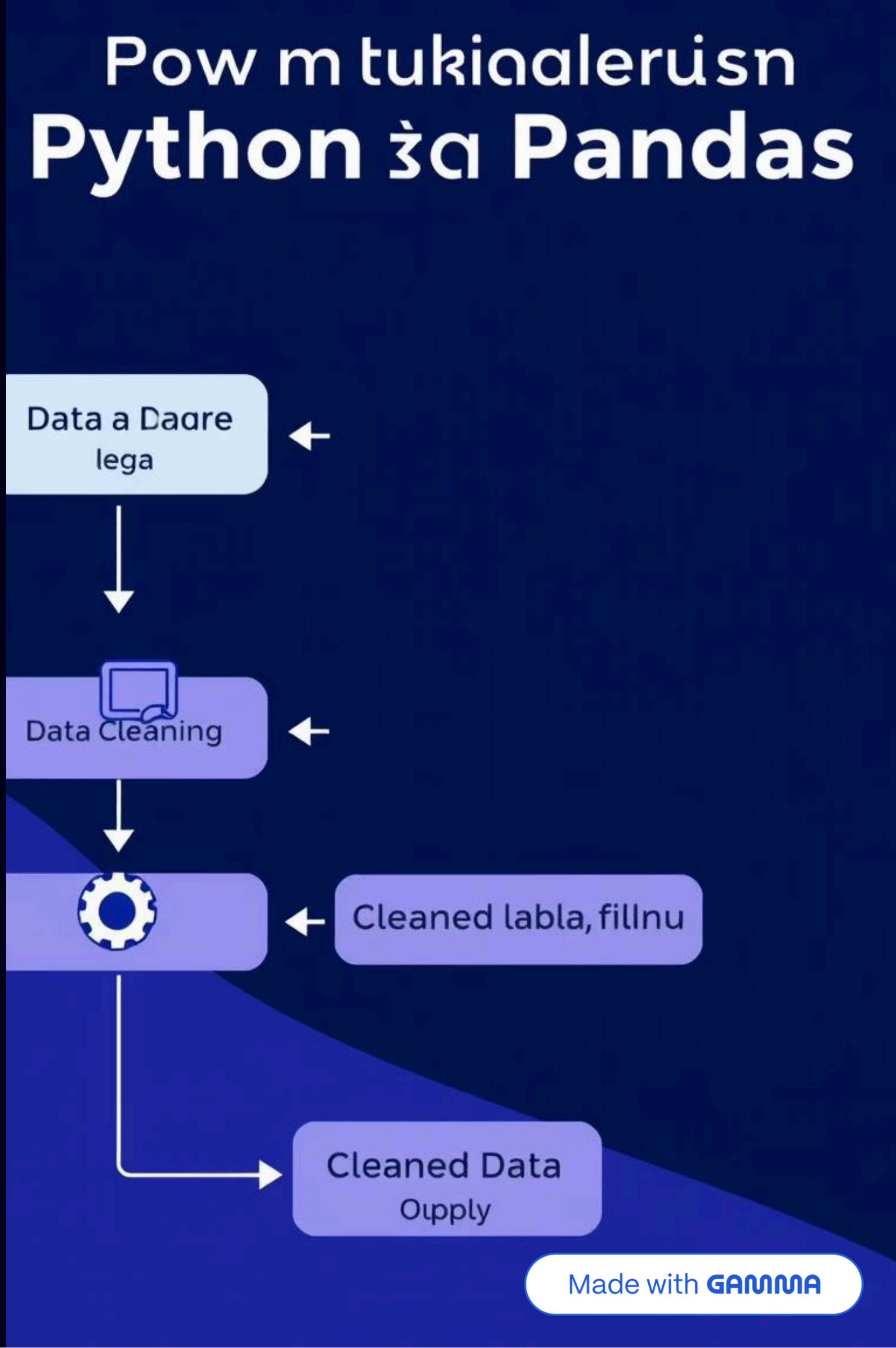
Convert data types

Merging & Joining

Combine data sets

Time Series

Analyze temporal data

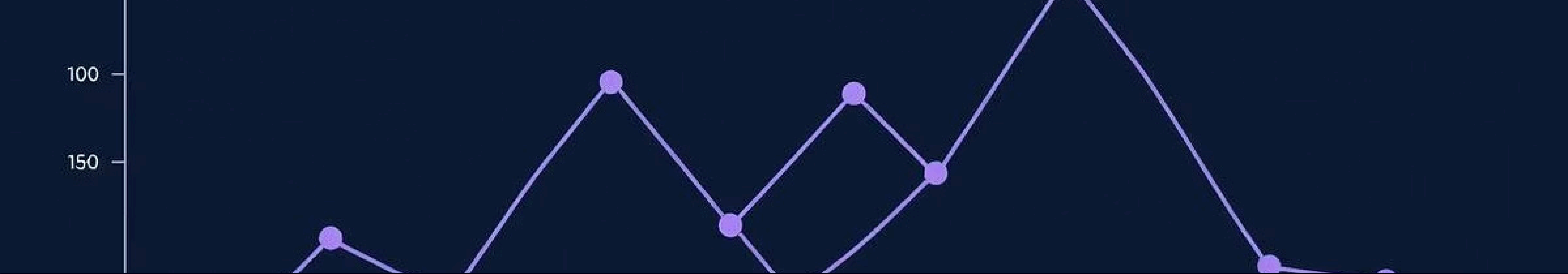


Applications of Pandas

Image Analysis with Pandas

Pandas can be integrated with image processing libraries to analyze image data effectively.

- Load image pixel data into a Pandas DataFrame for analysis.
- Perform statistical analysis on pixel intensities and color distributions.
- Use Pandas for image segmentation and feature extraction tasks.

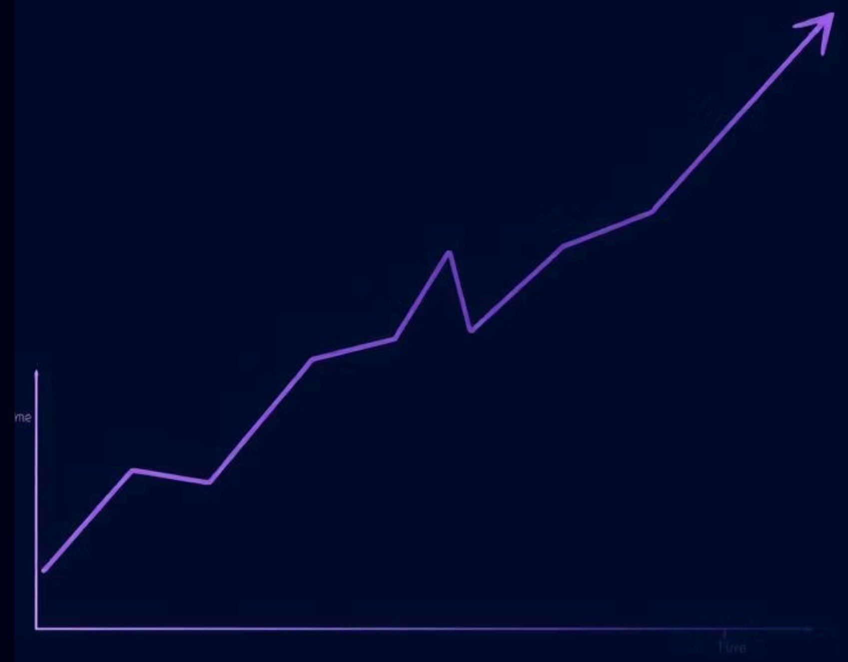


Matplotlib Overview

1. **Data Visualization**
   Create charts and plots
2. **Python Library**
   Essential visualization tool
3. **Graphs, Charts, Plots**
   Versatile output options

Matplotlib Features

- **Diverse Plotting:** Generate line, scatter, bar charts, histograms, and more.
- **Customization:** Fine-tune aesthetics with labels, colors, and styles.
- **Interactive Plots:** Create interactive figures with zoom and pan.
- **Export Options:** Save plots in various formats like PNG, JPG, PDF, and SVG.

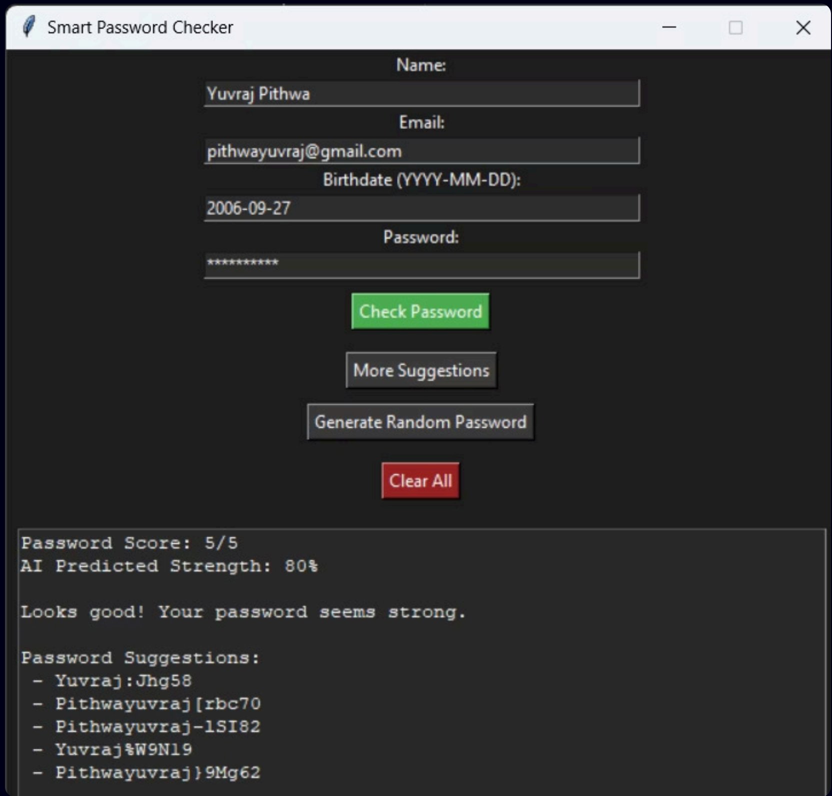


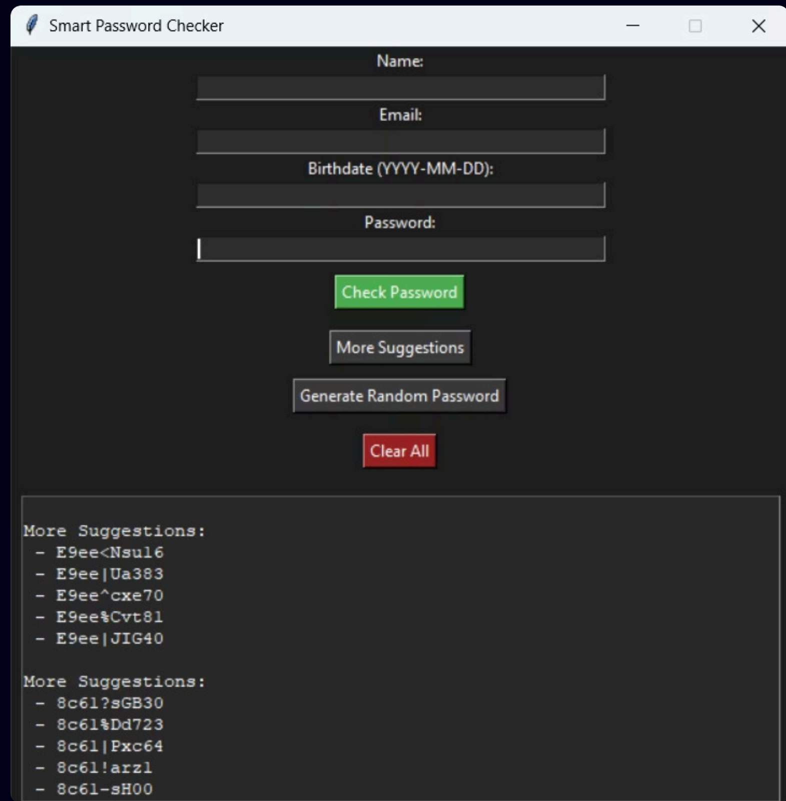
Applications of Matplotlib

Matplotlib is a versatile tool for visualizing image data. Here are some key applications:

- **Displaying Images:** Use Matplotlib to display images directly from NumPy arrays.
- **Visualizing Pixel Intensities:** Create histograms to analyze pixel intensity distributions.
- **Overlaying Plots on Images:** Combine image data with plots for detailed analysis.

Output of Project





Thank You

I covered a lot of ground in this presentation on popular Python data science libraries. From the fundamentals of NumPy and Pandas to the visualization capabilities of Matplotlib, we now have a solid understanding of the key tools used by data scientists and analysts.

These libraries provide powerful functionality for working with numerical data, tabular data, and creating visual representations. With the knowledge you've gained, you'll be able to start applying these tools to your own data analysis and projects.

Appreciating your efforts for your attention and participation throughout this presentation. I'm happy to answer any final questions you may have.